

## Exercise - 2

### 1. Sampling from exponential distribution

- (a) Derive the quantile function for exponential distribution ( $f(x) = \lambda \exp(-\lambda x)$ ,  $x \geq 0$ ) using its CDF. Using any programming language, sample from an exponential distribution (with rate parameter  $\lambda = 5$ ) on the basis of continuous uniform samples ( $p \sim \text{Uniform}(0, 1)$ ) utilising quantile function.
- (b) Learn about (via web search) the relationship between the Poisson and exponential distribution.
- (c) Apply what you learned in (b) to sample from a Poisson distribution (with mean parameter 5) using samples from the exponential distribution using any programming language.

### 2. Importance Sampling

Estimate the tail probability of  $\theta = P(X > 2)$  where  $X$  is Cauchy distributed (learn about Cauchy distribution via web search).

- (a) Use regular Monte Carlo sampling.
- (b) Use importance sampling with a standard-normal distribution as the proposal distribution.
- (c) Use importance sampling with a truncated Cauchy distribution as the proposal distribution.
- (d) Compare results obtained in (a), (b), and (c) to the true value.